Physics Numerical Problem Book

Theory-based numerical problems with step-by-step solutions

Chapter 1: Kinematics — Motion in a Straight Line

1.1 Theory

Kinematics is the branch of physics that describes the motion of objects without considering the forces that cause the motion. The key quantities are displacement, velocity, acceleration, and time.

Displacement (s): The change in position of an object. It is a vector quantity measured in metres (m).

Velocity (v): The rate of change of displacement. Measured in metres per second (m/s).

Acceleration (a): The rate of change of velocity. Measured in metres per second squared (m/s²).

Key Equations of Motion (SUVAT)

The following equations apply to uniform (constant) acceleration:

$$v = u + at$$

$$s = ut + (1/2)at^2$$

$$v^2 = u^2 + 2as$$

$$s = (u + v)/2 \times t$$

Where: u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement

1.2 Numerical Problems

Problem 1: A car starts from rest and accelerates uniformly at 4 m/s² for 6 seconds. Find its final velocity and the distance travelled.

Given: u = 0 m/s (starts from rest), a = 4 m/s², t = 6 s

Step 1 — Find final velocity using $v = u + at$

$v = 0 + (4)(6) = 24$ m/s

Step 2 — Find distance using $s = ut + (1/2)at^2$

$s = (0)(6) + (1/2)(4)(6^2) = 0 + 72 = 72$ m

Answer: Final velocity = 24 m/s, Distance = 72 m

Problem 2: A ball is thrown upward with an initial velocity of 20 m/s. How high does it go? ($g = 10$ m/s²)

Given: $u = 20$ m/s, $v = 0$ (at highest point), $a = -10$ m/s²

Using $v^2 = u^2 + 2as$

$0 = (20)^2 + 2(-10)(s)$

$0 = 400 - 20s$

$s = 400/20 = 20$ m

Answer: The ball reaches a maximum height of 20 m

Problem 3: A train moving at 72 km/h applies brakes and stops in 10 seconds. Find the deceleration and stopping distance.

Given: $u = 72$ km/h $= 20$ m/s, $v = 0$, $t = 10$ s

Step 1 — Find deceleration using $v = u + at$

$0 = 20 + a(10) \rightarrow a = -2$ m/s² (deceleration = 2 m/s²)

Step 2 — Find stopping distance using $s = (u+v)/2 \times t$

$s = (20+0)/2 \times 10 = 10 \times 10 = 100$ m

Answer: Deceleration = 2 m/s², Stopping distance = 100 m

Chapter 2: Newton's Laws of Motion

2.1 Theory

Newton's First Law (Law of Inertia): An object remains at rest or in uniform motion unless acted upon by an external net force.

Newton's Second Law: The net force acting on an object equals the product of its mass and acceleration.

$$F = m \times a$$

Where: F = net force (Newtons, N), m = mass (kg), a = acceleration (m/s²)

Newton's Third Law: For every action there is an equal and opposite reaction.

Weight: The gravitational force on an object. W = mg, where g = 9.8 m/s² (≈ 10 m/s² for simplicity).

$$W = m \times g$$

Friction Force: The force that opposes motion between two surfaces in contact.

$$f = \mu \times N$$

Where: μ = coefficient of friction, N = normal force

2.2 Numerical Problems

Problem 4: A 10 kg box is pushed with a net force of 50 N. What is its acceleration?

Given: m = 10 kg, F = 50 N

Using F = ma $\rightarrow$ a = F/m

a = 50/10 = 5 m/s²

Answer: Acceleration = 5 m/s²

Problem 5: A 5 kg object rests on a surface with coefficient of friction μ = 0.3. What is the friction force? (g = 10 m/s²)

Given: m = 5 kg, μ = 0.3, g = 10 m/s²

Step 1 — Normal force N = mg = 5 × 10 = 50 N

Step 2 — Friction force f = μN = 0.3 × 50 = 15 N

Answer: Friction force = 15 N

Problem 6: A 1200 kg car accelerates from 0 to 20 m/s in 8 seconds. What force does the engine apply?

Given: m = 1200 kg, u = 0, v = 20 m/s, t = 8 s

Step 1 — Find acceleration: a = (v-u)/t = (20-0)/8 = 2.5 m/s²

Step 2 — Find force: F = ma = 1200 × 2.5 = 3000 N

Answer: Engine force = 3000 N

Chapter 3: Work, Energy and Power

3.1 Theory

Work (W): Work is done when a force moves an object through a displacement. Work is measured in Joules (J).

```
W = F × d × cos(θ)
```

Where: F = force (N), d = displacement (m), θ = angle between force and displacement. If force is parallel to motion, θ = 0 and cos(θ) = 1, so W = F × d.

Kinetic Energy (KE): The energy possessed by an object due to its motion.

```
KE = (1/2) × m × v²
```

Potential Energy (PE): The energy possessed by an object due to its position or height.

```
PE = m × g × h
```

Where: h = height above reference point (m)

Law of Conservation of Energy: Energy cannot be created or destroyed. It can only be converted from one form to another. Total mechanical energy = KE + PE = constant.

Power (P): The rate at which work is done. Measured in Watts (W).

```
P = W/t or P = F × v
```

3.2 Numerical Problems

Problem 7: A person pushes a box with a force of 100 N over a distance of 15 m. How much work is done?

Given: F = 100 N, d = 15 m, $\theta = 0°$

$W = F \times d = 100 \times 15 = 1500$ J

Answer: Work done = 1500 Joules

Problem 8: A 3 kg ball moves at 10 m/s. What is its kinetic energy?

Given: m = 3 kg, v = 10 m/s

$KE = (1/2) \times m \times v^2 = (1/2) \times 3 \times (10)^2 = (1/2) \times 3 \times 100 = 150$ J

Answer: Kinetic energy = 150 Joules

Problem 9: A 2 kg book is placed on a shelf 3 m above the ground. What is its potential energy? (g = 10 m/s²)

Given: m = 2 kg, h = 3 m, g = 10 m/s²

$PE = mgh = 2 \times 10 \times 3 = 60$ J

Answer: Potential energy = 60 Joules

Problem 10: A ball of mass 0.5 kg is dropped from a height of 20 m. What is its velocity just before hitting the ground? (g = 10 m/s²)

Given: m = 0.5 kg, h = 20 m, g = 10 m/s²

Using conservation of energy: PE at top = KE at bottom

$mgh = (1/2)mv^2$

$v^2 = 2gh = 2 \times 10 \times 20 = 400$

v = 20 m/s

Answer: Velocity just before impact = 20 m/s

Problem 11: A machine does 6000 J of work in 2 minutes. What is its power output?

Given: W = 6000 J, t = 2 min = 120 s

P = W/t = 6000/120 = 50 W

Answer: Power = 50 Watts

Chapter 4: Electricity — Ohm's Law and Circuits

4.1 Theory

Electric Current (I): The flow of electric charge. Measured in Amperes (A).

```
I = Q/t
```

Where: Q = charge in Coulombs (C), t = time in seconds (s)

Voltage (V): The potential difference between two points. Measured in Volts (V). It is the energy per unit charge.

Resistance (R): The opposition to current flow. Measured in Ohms (Ω).

Ohm's Law: The current through a conductor is directly proportional to the voltage across it (at constant temperature).

```
V = I × R
```

Electrical Power: The rate at which electrical energy is consumed.

```
P = V × I = I² × R = V²/R
```

Resistors in Series: Total resistance is the sum of individual resistances.

```
R_total = R1 + R2 + R3 + ...
```

Resistors in Parallel: The reciprocal of total resistance equals the sum of reciprocals.

```
1/R_total = 1/R1 + 1/R2 + 1/R3 + ...
```

4.2 Numerical Problems

Problem 12: A circuit has a voltage of 12 V and a resistance of 4 Ω. What is the current?

Given: V = 12 V, R = 4 Ω

Using V = IR → I = V/R = 12/4 = 3 A

Answer: Current = 3 Amperes

Problem 13: A bulb has a resistance of 240 Ω and is connected to a 120 V supply. Find the current and power consumed.

Given: R = 240 Ω, V = 120 V

Step 1 — Current: I = V/R = 120/240 = 0.5 A

Step 2 — Power: P = V × I = 120 × 0.5 = 60 W

Answer: Current = 0.5 A, Power = 60 Watts

Problem 14: Two resistors of 6 Ω and 12 Ω are connected in parallel. What is the total resistance?

Given: R1 = 6 Ω, R2 = 12 Ω

1/R_total = 1/6 + 1/12 = 2/12 + 1/12 = 3/12 = 1/4

R_total = 4 Ω

Answer: Total resistance = 4 Ohms

Problem 15: Three resistors of 2 Ω, 3 Ω, and 5 Ω are connected in series to a 20 V battery. Find total resistance and current.

Given: R1 = 2 Ω, R2 = 3 Ω, R3 = 5 Ω, V = 20 V

R_total = R1 + R2 + R3 = 2 + 3 + 5 = 10 Ω

I = V/R = 20/10 = 2 A

Answer: Total resistance = 10 Ω, Current = 2 A

Chapter 5: Waves and Sound

5.1 Theory

A wave is a disturbance that transfers energy through a medium without transferring matter.

Frequency (f): The number of complete waves passing a point per second. Measured in Hertz (Hz).

Wavelength (λ): The distance between two consecutive crests or troughs. Measured in metres (m).

Wave Speed (v): The speed at which a wave travels through a medium.

```
v = f × λ
```

Period (T): The time taken for one complete wave. T = 1/f

```
T = 1/f
```

Speed of Sound: Sound travels at approximately 340 m/s in air at room temperature.

Speed of Light: Light travels at 3 × 10■ m/s in a vacuum.

Echo: An echo is a reflection of sound. The distance to a reflecting surface can be calculated using:

```
d = (v × t) / 2
```

Where d is distance to the wall, v is speed of sound, and t is total time for sound to travel to the wall and back.

5.2 Numerical Problems

Problem 16: A wave has a frequency of 500 Hz and a wavelength of 0.68 m. What is its speed?

Given: f = 500 Hz, λ = 0.68 m

v = f × λ = 500 × 0.68 = 340 m/s

Answer: Wave speed = 340 m/s (speed of sound in air)

Problem 17: A sound wave travels at 340 m/s. If its wavelength is 0.085 m, what is its frequency?

Given: v = 340 m/s, λ = 0.085 m

f = v/λ = 340/0.085 = 4000 Hz

Answer: Frequency = 4000 Hz = 4 kHz

Problem 18: A person claps near a wall and hears the echo after 0.5 seconds. How far away is the wall? (speed of sound = 340 m/s)

Given: t = 0.5 s, v = 340 m/s

d = (v × t) / 2 = (340 × 0.5) / 2 = 170 / 2 = 85 m

Answer: The wall is 85 metres away

Problem 19: What is the period of a wave with frequency 50 Hz?

Given: f = 50 Hz

T = 1/f = 1/50 = 0.02 seconds

Answer: Period = 0.02 s = 20 milliseconds